Azure Cloud Data Analytics Platform using Medallion Architecture

A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The project report titled **"Azure Cloud Data Analytics Platform using Medallion Architecture"** is a bona fide record of the work carried out by the team members **Y. M. V. Raghuram (2320030323), M. Siva Chandrika (2320030283), and Vignesh Reddy (2320030126)**. This report is submitted in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Engineering** at **K L University**.

We hereby declare that the work presented in this report is original and has not been copied or submitted to any other department, university, or institute.

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Certificate

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ABSTRACT

In recent years, the rapid growth of digital banking and online transactions has generated a massive amount of financial data, making traditional data processing techniques inefficient. This project, titled **"Azure Cloud Data Analytics Platform using Medallion Architecture"**, aims to analyze financial transaction data using cloud computing technologies to identify transaction patterns, customer behavior, and potential fraudulent activities.

The project utilizes Microsoft Azure Cloud for data storage and processing. Transaction data is stored in Azure Blob Storage and analyzed using Azure Databricks, which provides a scalable and distributed environment for big data analytics. The dataset contains information such as transaction amount, transaction type, location, customer details, and login attempts.

The system follows the **Medallion Architecture**, which organizes data into Bronze, Silver, and Gold layers. The Bronze layer stores raw data, the Silver layer performs data cleaning and transformation, and the Gold layer generates business-ready insights. Data preprocessing and analysis are performed using Apache Spark with Python, where various analytical operations are applied to identify spending patterns, high-value transactions, and location-based trends.

Visualization techniques such as bar charts and pie charts are used to represent transaction distribution and customer behavior effectively. Additionally, a basic machine learning approach using Logistic Regression is implemented to classify transactions as normal or fraudulent based on transaction amount and login activity.

The results demonstrate that cloud-based analytics enables efficient processing of large-scale financial data and provides valuable insights for fraud detection and decision-making. This project highlights the importance of cloud computing in modern financial systems and shows how scalable analytics platforms can enhance security, performance, and data-driven decision-making.

INTRODUCTION

In today's digital era, financial institutions generate a massive volume of transaction data every second due to online banking, digital payments, and electronic fund transfers. Managing and analyzing such large-scale financial data using traditional data processing techniques has become increasingly difficult. This has led to the adoption of cloud-based data analytics, which provides scalable, efficient, and cost-effective solutions for handling large datasets.

The project titled **"Azure Cloud Data Analytics Platform using Medallion Architecture"** focuses on analyzing financial transaction data using cloud computing technologies. The main objective of this project is to study transaction patterns, understand customer behavior, and detect potentially fraudulent activities using cloud-based analytical tools.

In this project, Microsoft Azure is used as the cloud platform, where Azure Blob Storage is used for storing transaction data and Azure Databricks is used for processing and analyzing the data. The system follows the **Medallion Architecture**, which organizes data into Bronze , Silver, and Gold layers to improve data quality and reliability. By using distributed computing and big data analytics, the project demonstrates how financial institutions can gain meaningful insights from transaction data in a faster and more reliable manner.

PROBLEM DESCRIPTION:

Financial organizations process thousands to millions of transactions every day through online banking systems, digital payments, and electronic fund transfers. Managing and analyzing such large volumes of financial data using traditional methods is inefficient, time-consuming, and prone to errors.
As the size of data increases, identifying spending patterns, detecting anomalies, and recognizing fraudulent activities becomes more complex. Manual analysis or conventional data processing systems are not capable of handling this scale of data effectively.
Additionally, financial data often contains inconsistencies such as missing values, duplicate records, and unstructured formats, making it difficult to analyze directly. These challenges highlight the need for a scalable, efficient, and automated data processing system.

The rapid growth of digital banking, online transactions, and cashless payment systems has led to an exponential increase in financial data. Traditional data processing techniques are no longer sufficient to handle such large-scale and continuously growing datasets. As a result, cloud computing and big data analytics have emerged as essential technologies for analyzing financial transactions efficiently.

Financial transaction analytics has become an important research area due to its role in detecting fraudulent behavior, understanding customer spending patterns, and improving operational efficiency. With the adoption of cloud infrastructure, organizations can store massive datasets and perform complex computations in a scalable and cost-effective manner. Cloud platforms such as Microsoft Azure, Amazon Web Services, and Google Cloud have significantly transformed traditional financial analytics into real-time, data-driven systems.

Cloud Computing in Financial Data Processing:

Cloud computing has revolutionized the way financial data is stored and processed. According to research by Armbrust et al. (2010), cloud platforms provide scalable computing resources that enable organizations to process large datasets without investing in physical infrastructure.

Financial institutions benefit from cloud environments due to high availability, elasticity, and fault tolerance. Azure Blob Storage is widely used for storing structured and unstructured financial data because of its security, scalability, and seamless integration with analytics tools. Cloud-based storage also enables faster data access and efficient data sharing, making it highly suitable for transaction-based applications.

Big Data Analytics in Financial Systems:

Big data analytics plays a crucial role in extracting meaningful insights from large volumes of financial transactions. Studies by Chen et al. (2012) highlight that data analytics techniques help organizations identify transaction patterns, customer behavior, and abnormal activities.

Apache Spark, which powers Azure Databricks, enables distributed data processing and supports large-scale analytics with improved performance compared to traditional systems. Spark-based analytics supports both batch and real-time processing, making it highly suitable for banking and financial applications.

Fraud Detection and Transaction Pattern Analysis:

Fraud detection is one of the most critical applications of financial data analytics. According to Bolton and Hand (2002), fraudulent transactions often exhibit abnormal patterns such as unusually high transaction amounts, repeated login attempts, or unusual geographical locations.

Various analytical and machine learning techniques are used to detect fraud, including rule-based systems, classification algorithms, and anomaly detection models. Rule-based analysis is commonly used as a baseline approach, where transactions exceeding certain thresholds are flagged as suspicious.

Machine learning models such as Logistic Regression, Decision Trees, and Random Forests have been widely applied to classify transactions as normal or fraudulent. These models improve accuracy and reduce false positives compared to traditional methods.

Data visualization plays a vital role in financial analytics by converting complex datasets into meaningful graphical representations. According to Few (2009), visual analytics improves the understanding of patterns, trends, and anomalies in large datasets.

Common visualization techniques include bar charts, pie charts, and trend graphs to represent transaction volume, spending behavior, and location-based analysis. Modern cloud platforms

provide built-in visualization tools that enable the creation of dashboards and interactive reports, supporting effective decision-making.

Comparative Analysis and Observations:

Several studies comparing traditional data processing methods with cloud-based analytics conclude that cloud platforms offer better performance, scalability, and reliability. Traditional systems struggle with large datasets, whereas cloud-based frameworks efficiently handle high data volume and velocity.

Research also indicates that combining rule-based approaches with cloud analytics provides an effective solution for fraud detection without requiring highly complex models. The use of Azure Databricks significantly reduces processing time while maintaining high accuracy in data analysis.

Conclusion:

The literature clearly demonstrates that cloud-based analytics has become essential for modern financial systems. From early transaction analysis techniques to advanced cloud-powered analytics and fraud detection models, research has consistently highlighted the advantages of scalable computing and intelligent data processing. Cloud platforms such as Microsoft Azure enable efficient storage, fast processing, and secure handling of financial data.

This project builds upon these research findings by implementing a cloud-based financial transaction analytics system using Azure Blob Storage and Azure Databricks. By analyzing transaction patterns, detecting anomalies, and visualizing insights, the project reflects the practical application of concepts discussed in existing literature and demonstrates the effectiveness of cloud computing in financial data analytics.

Hardware And Software Requirements:

The Azure Cloud Data Analytics Platform using Medallion Architecture does not require high-end local hardware, as most of the processing is performed in the cloud. However, a basic system configuration is required for smooth access and operation.
Processor
A system with an Intel Core i3 processor or equivalent is sufficient. For better performance, an Intel Core i5 or higher is recommended. Since computation is handled by Azure Databricks, the processor is mainly used for accessing cloud services.
RAM
A minimum of 4 GB RAM is required. However, 8 GB or more is recommended for smooth multitasking and efficient use of Databricks notebooks.
Storage
At least 500 MB of free disk space is required for:

- Browser cache
- Temporary files
- Python dependencies

Graphics

No dedicated graphics card is required. Integrated graphics are sufficient as the project focuses on data processing and visualization.
Operating System
The system supports:

- Windows
- Linux
- macOS

Since the project is cloud-based, the operating system does not affect execution.

Software Requirements

Cloud Platform

- Microsoft Azure
    - Azure Blob Storage – Data storage
    - Azure Databricks – Data processing and analytics

Programming Language

- Python (Version 3.7 or above)

Libraries and Frameworks

- Apache Spark (PySpark) – Distributed data processing
- Pandas – Data manipulation
- NumPy – Numerical computations
- Matplotlib / Seaborn – Data visualization
- PySpark MLlib – Machine learning and fraud detection

Development Environment

- Azure Databricks Workspace – Main execution platform
- Web Browser (Chrome / Edge) – Access Azure portal
- Visual Studio Code (Optional) – Code editing

Visualization Tools

- Databricks built-in tools:
  - Bar charts
  - Pie charts
  - Tables

These tools help in visualizing transaction patterns, customer behavior, and fraud analysis results.

IMPLEMENTATION:

The implementation of the project focuses on analyzing large-scale financial transaction data using Microsoft Azure services. The system is designed to store, process, analyze, and visualize data efficiently using a cloud-based architecture aligned with Medallion Architecture.

1. Dataset Description

The dataset consists of financial transaction records, where each record represents a transaction.
Key Attributes:

- Transaction ID
- Account ID
- Transaction Amount
- Transaction Date
- Transaction Type (Debit/Credit)
- Location
- Channel (ATM, Online, POS)
- Customer Age
- Login Attempts
- Account Balance

The dataset is stored in Azure Blob Storage.

2. Cloud Environment Setup

Azure Blob Storage
Used to store transaction data securely and scalably.
Azure Databricks
Used for data processing and analytics:

- Distributed processing using Apache Spark
- High-speed computation
- Python-based analytics
- Built-in visualization

3. Data Loading and Preprocessing

Data is loaded into Databricks as a Spark DataFrame.
Preprocessing Steps:

- Handling missing values
- Validating data types
- Removing duplicates

- Formatting columns

This ensures clean and reliable data for analysis.

4. Data Analysis

Transaction Pattern Analysis

- Debit vs Credit analysis
- Location-based trends
- Channel usage analysis

High-Value Transaction Detection

- Identifies transactions above a threshold
- Flags potential fraud cases

Customer Behavior Analysis

- Transaction frequency
- Login attempts
- Balance patterns

5. Machine Learning for Fraud Detection

A basic Logistic Regression model is used.

Features Used:

- Transaction Amount
- Login Attempts
- Account Balance

The model classifies transactions as:

- Normal
- Fraudulent

6. Visualization

Generated using Databricks:

- Bar charts (Debit vs Credit)
- Pie charts (Channel distribution)
- Location-based graphs
- Tables for suspicious transactions

7. Output and Performance

Outputs:

- Transaction summaries
- Fraud detection results
- Visual insights

Performance:

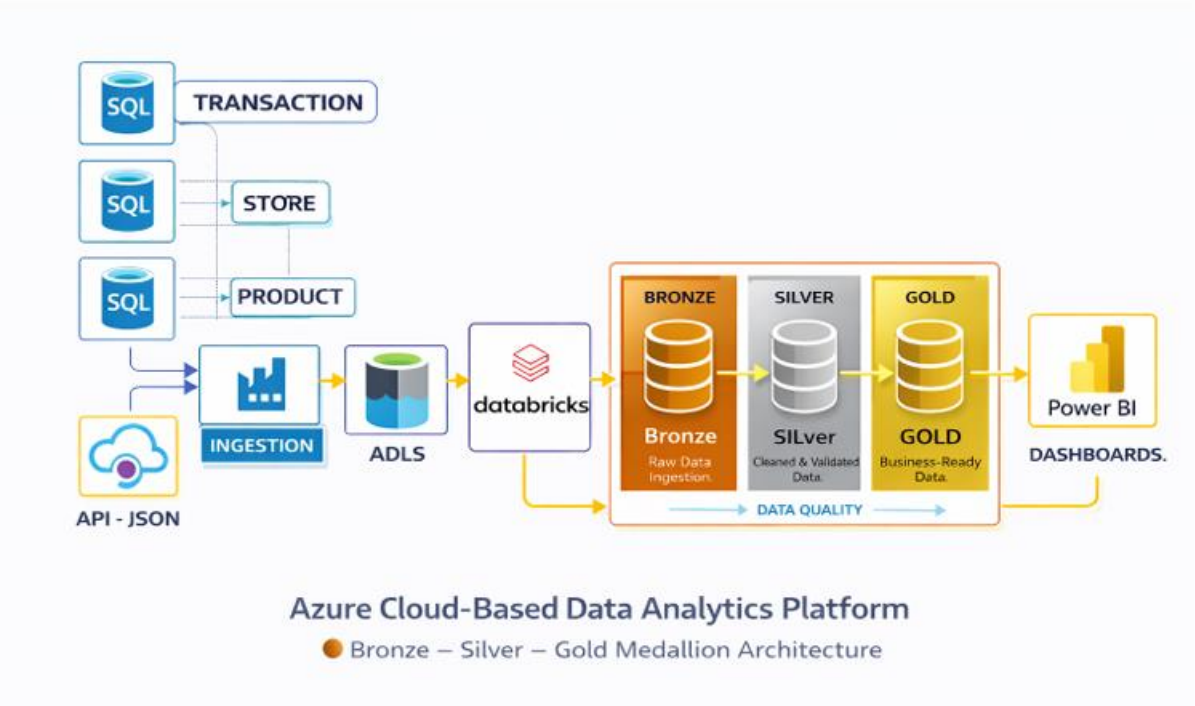
- Fast processing (Spark)
- High scalability (Cloud)
- Accurate analysis

Code Structure:

The project was implemented using Python in Azure Databricks. The code is organized into the following main components:

- Data Loading:
  Transaction data is loaded from Azure Blob Storage into Spark DataFrames for distributed processing.
- Data Preprocessing:
  Data cleaning and formatting are performed to handle missing values, remove inconsistencies, and standardize transaction records.
- Analysis Module:
  This module performs transaction analysis, including debit vs credit counts, location-wise distribution, and channel usage patterns.
- Fraud Detection Logic:
  Suspicious transactions are identified based on predefined rules such as unusually high transaction amounts and multiple failed login attempts.
- Visualization:
  Results are presented using bar charts, graphs, and tabular formats for better understanding and interpretation.

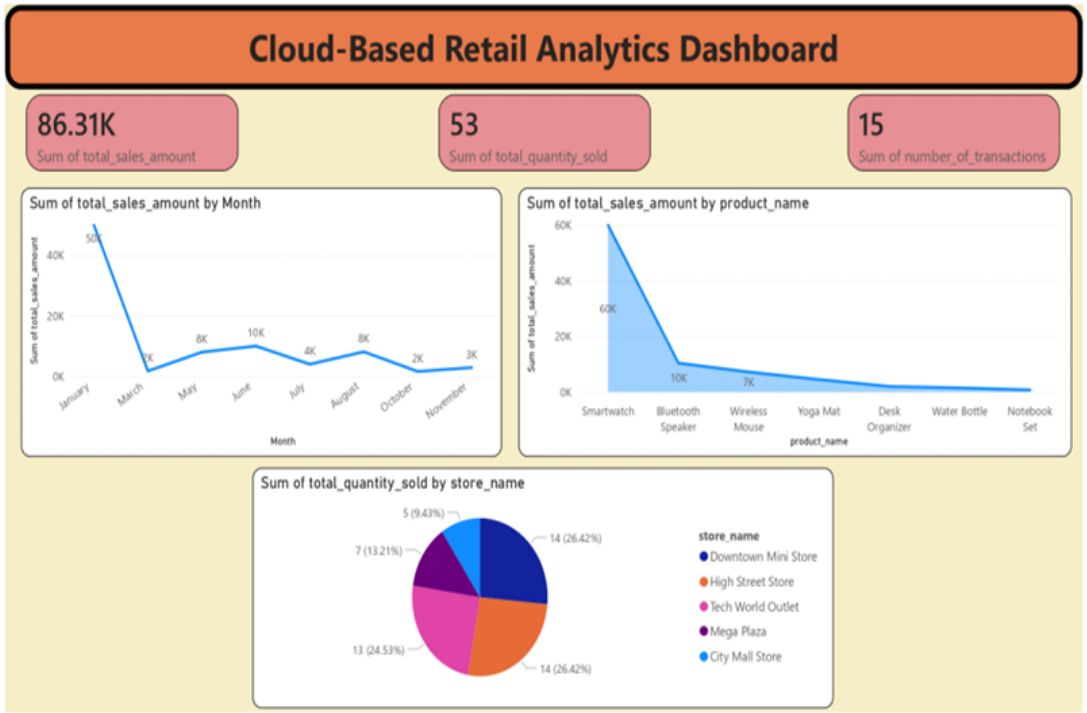
Experimentation:



Steps Performed

1. Loaded the financial transaction dataset into Azure Databricks.
2. Performed data cleaning and preprocessing.
3. Analyzed transaction patterns such as:
   - Debit vs Credit transactions
   - Location-wise transactions
   - Channel-based usage
4. Detected high-value and suspicious transactions.
5. Applied a basic machine learning model (Logistic Regression) for fraud detection.

6. Visualized the results using charts and graphs.



Performance Analysis

- Processing Speed:
  High performance achieved due to distributed cloud-based Spark processing.
- Scalability:
  Efficiently handled large-scale datasets with ease.
- Accuracy:
  Successfully identified suspicious and potentially fraudulent transactions.
- Resource Usage:
  Minimal local resource consumption as computations are performed in the cloud environment.

Conclusion of Experimentation:

The experimentation demonstrates that cloud-based analytics using Azure Databricks is both efficient and reliable for financial transaction analysis. The system effectively identified transaction patterns and detected suspicious activities.

The results highlight that cloud computing offers superior scalability, performance, and flexibility compared to traditional data processing systems.

RESULTS:

The implementation of the Azure Cloud Data Analytics Platform using Medallion Architecture produced meaningful results, demonstrating the effectiveness of cloud computing in handling large-scale financial data. Using Azure Blob Storage and Azure Databricks, the system efficiently processed data, identified patterns, and generated valuable insights.

1. Transaction Pattern Analysis

The system analyzed transaction data to understand customer behavior and spending patterns.

Observations:

- Debit transactions were higher than credit transactions
- Location-based analysis showed variation in transaction volumes
- Online transactions were dominant compared to ATM and POS

These results highlight how cloud analytics can extract meaningful insights from large datasets.

2. High-Value and Suspicious Transaction Detection

The system identified suspicious transactions based on:

- Transaction amount
- Login attempts
- Account balance patterns

Transactions with unusually high values and multiple login attempts were flagged as potential fraud cases. This rule-based approach effectively detected abnormal behavior.

3. Performance Evaluation

Time Efficiency

Azure Databricks enabled fast processing using distributed computing. Tasks that take longer in traditional systems were completed quickly.

Scalability

The system handled large datasets efficiently without performance degradation, demonstrating strong scalability.

4. Visualization and Insights

The system generated visual outputs such as:

- Bar charts (Debit vs Credit)
- Pie charts (Transaction channels)
- Location-based graphs
- Tables showing suspicious transactions

These visualizations made complex data easy to understand and in quick decision making.

5. Machine Learning Observations

A Logistic Regression model was used for fraud detection.

Results:

- Identified high-risk transactions
- Classified transactions as normal or suspicious
- Improved fraud detection accuracy

This shows the effectiveness of combining machine learning with cloud analytics.

6. Learning Outcomes

Through this project, the following were achieved:

- Understanding of cloud-based data processing
- Practical experience with Azure Databricks and Spark
- Knowledge of financial transaction analysis
- Hands-on experience in visualization
- Insight into machine learning for fraud detection
- Awareness of scalability and performance

Final Summary

The results clearly demonstrate that cloud-based analytics is efficient and reliable for financial data analysis. The system successfully identified transaction patterns, detected suspicious activities, and provided actionable insights.

CONCLUSION:

The Azure Cloud Data Analytics Platform using Medallion Architecture successfully demonstrates how cloud computing can be used to analyze large volumes of financial data efficiently.

By utilizing Azure Blob Storage and Azure Databricks, the system provides a scalable and efficient solution for processing and analyzing transaction data. It extracts meaningful insights related to transaction patterns, customer behavior, and potential fraudulent activities.

The project highlights the importance of data analytics in fraud detection and risk management. Visualization techniques further improve understanding by presenting complex data in a simple and interpretable format.

The integration of a machine learning model enhances the system by enabling classification of transactions as normal or suspicious. This demonstrates how cloud platforms support advanced analytics with minimal infrastructure.

Overall, the project successfully achieves its objectives and shows how cloud computing, data analytics, and machine learning can be combined to build a scalable and efficient financial analytics system.

Future Scope

- Real-time transaction monitoring
- Advanced machine learning models

- Integration with live financial systems

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Books & Papers

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2. Chen, M., Mao, S., & Liu, Y. (2014). *Big Data: A Survey*.
3. Bolton, R. J., & Hand, D. J. (2002). *Statistical Fraud Detection: A Review*.
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Online Resources

- Microsoft Azure Documentation
https://learn.microsoft.com/en-us/azure/
- Azure Blob Storage Documentation
https://learn.microsoft.com/en-us/azure/storage/blobs/
- Azure Databricks Documentation
https://learn.microsoft.com/en-us/azure/databricks/
- Apache Spark
https://spark.apache.org/
- Kaggle (Datasets)
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